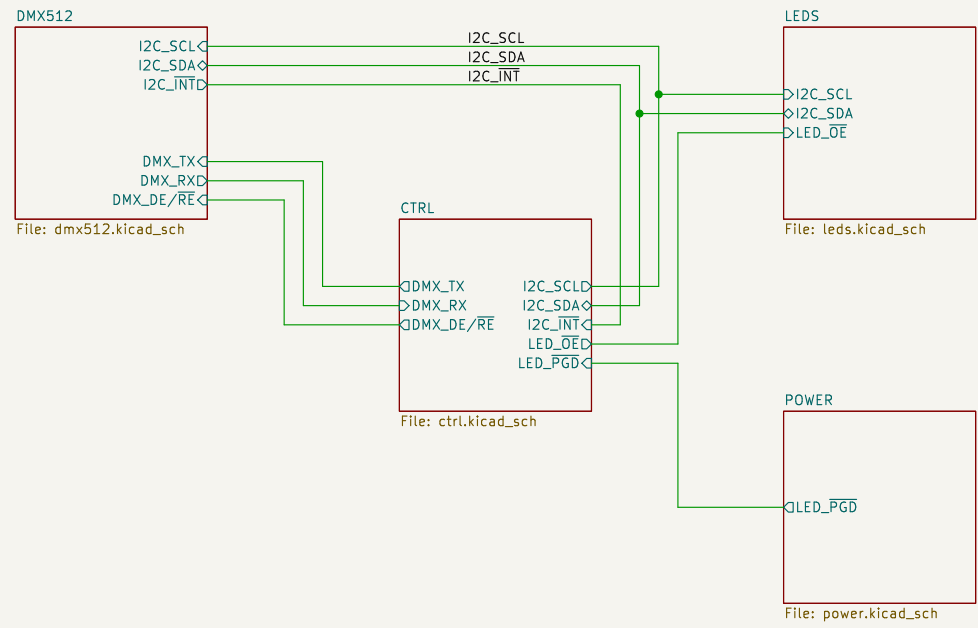


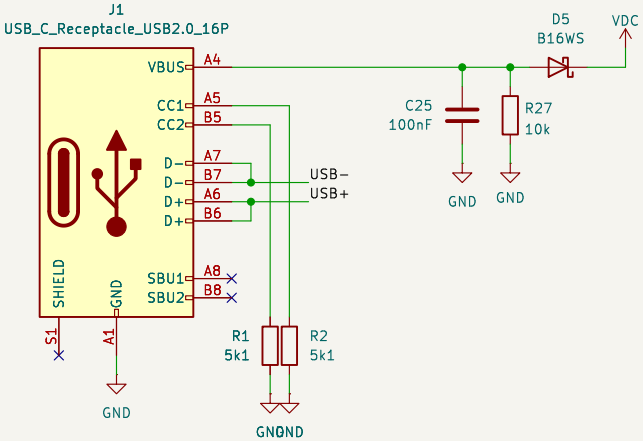
LED Beacon



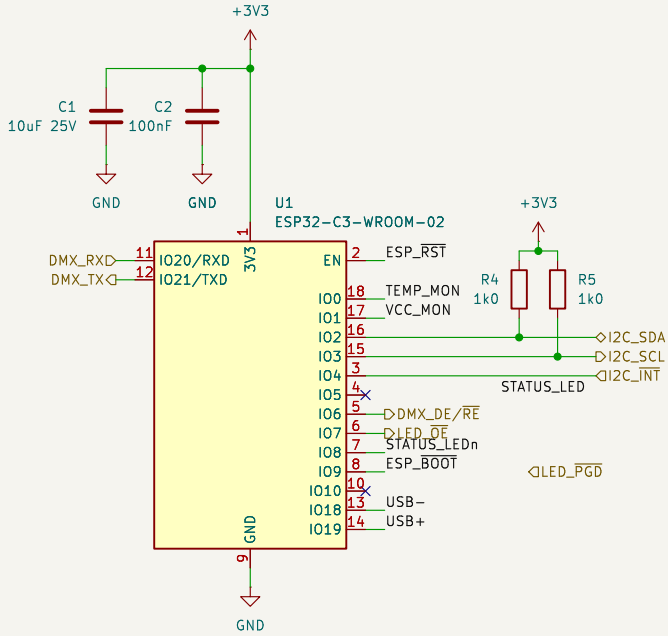
- H1 MountingHole
- H2 MountingHole
- H3 MountingHole
- H4 MountingHole

Wires & Bits		
Sheet: /		
File: emf_beacon.kicad_sch		
Title:		
Size: A4	Date: 2026-04-10	Rev: 0
KiCad E.D.A. 9.0.9		Id: 1/5

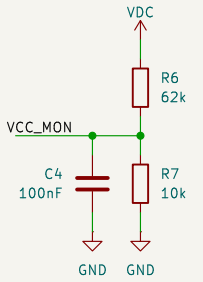
USB Programming



Controller

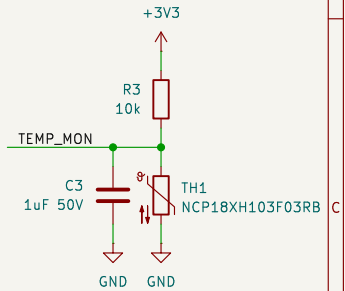


VCC Monitoring

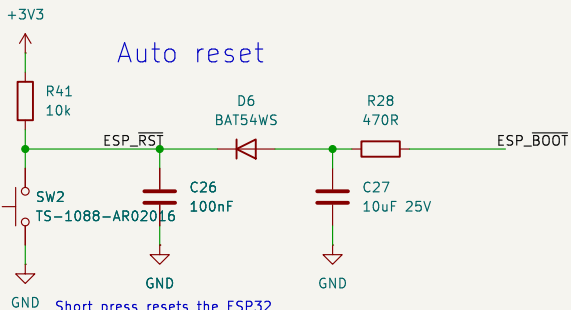


Voltage Monitor GND GND
Vmax is 17.5V
 $V_{adc} \times ((43k\Omega + 10k\Omega) \div 10k\Omega) = VCC$
 $V_{adc} \times 5.3 = VCC$
 $1lsb = 4,271\mu V$

Temperature Monitoring

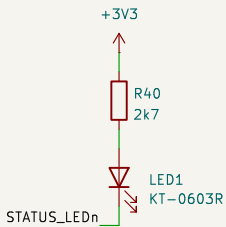


Auto reset



Short press resets the ESP32
Long press resets the ESP32
and puts it in bootloader mode

Status LED



IO8 needs to be held high to boot
so LED & resistor are high side.
Set STATUS_LEDn to 0V to light LED.

Wires & Bits

Sheet: /CTRL/
File: ctrl.kicad_sch

Title:

Size: A4	
KiCad E.D.A. 9.0.9	

Date: 2026-04-10

Rev: 0
Id: 2/5

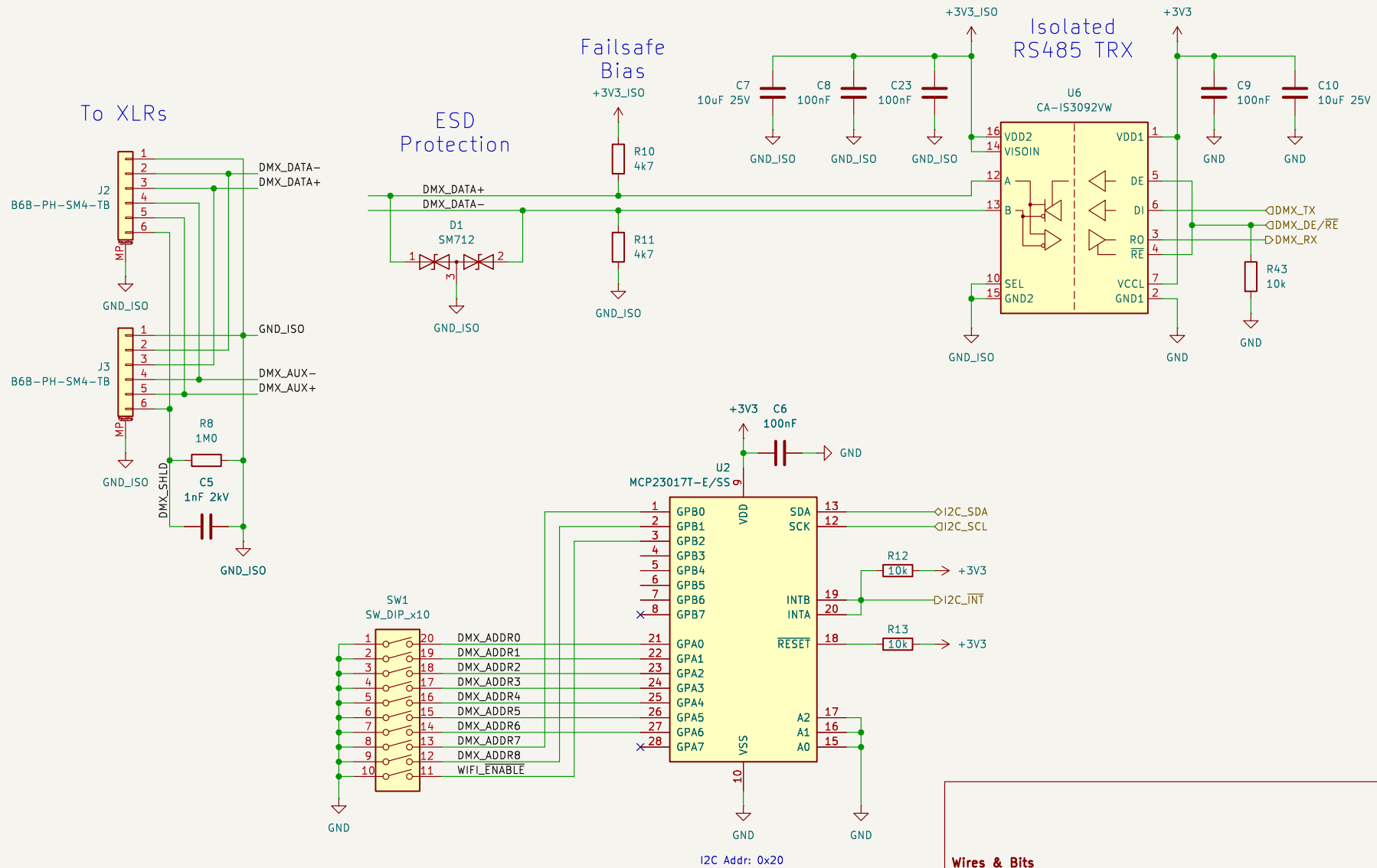
DMX512

Failsafe Bias

ESD Protection

Isolated RS485 TRX

To XLRs



Wires & Bits

Sheet: /DMX512/
File: dmx512.kicad_sch

Title:

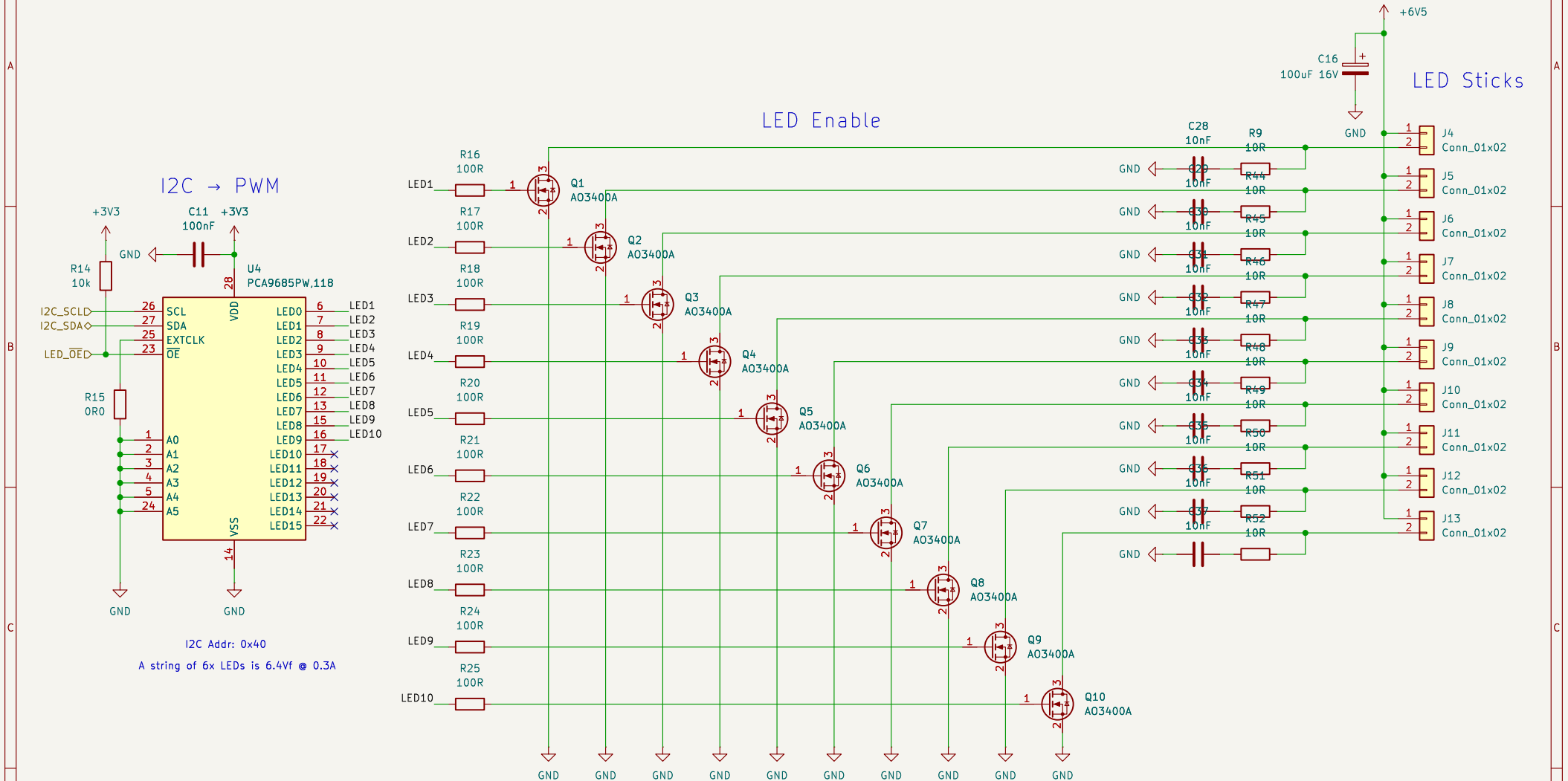
Size: A4
KiCad E.D.A. 9.0.9

Date: 2026-04-10

Rev: 0

Id: 3/5

LED Driver



Wires & Bits

Sheet: /LEDS/
File: leds.kicad_sch

Title:

Size: A4
KiCad E.D.A. 9.0.9

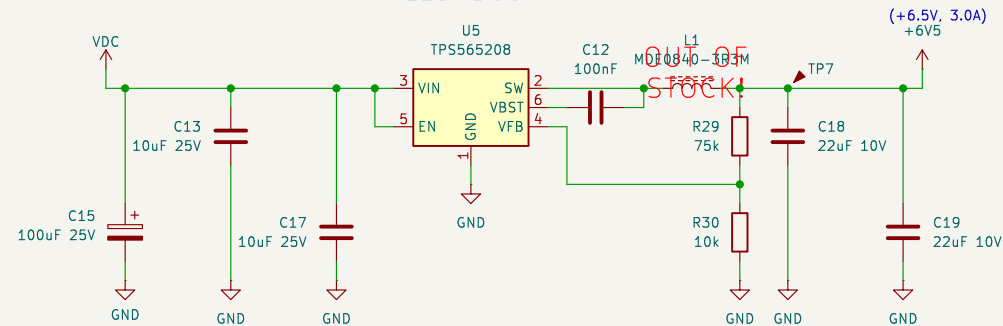
Date: 2026-04-10

Rev: 0

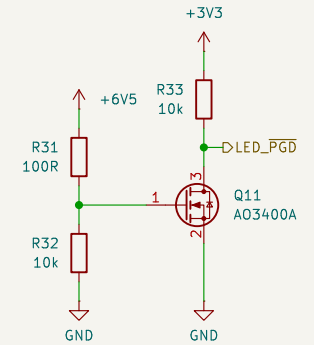
Id: 4/5

Power Supplies

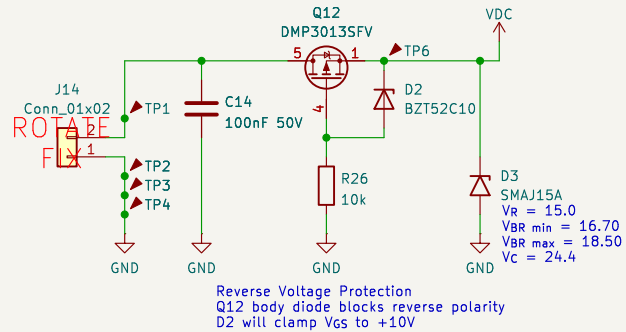
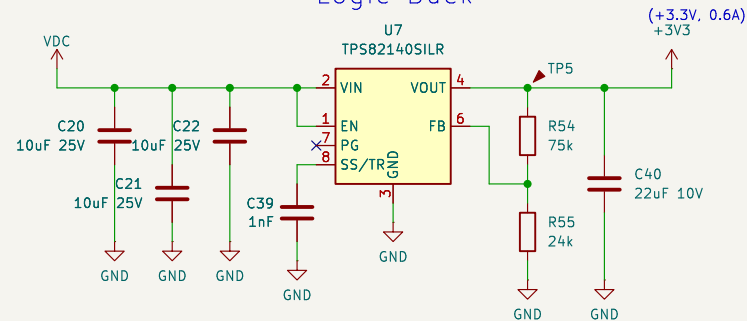
LED Buck



LED Power Good



Logic Buck



Find a good SMD Power input connector
lever terminal blocks?

EMC ferrites /CMC?

Wires & Bits

Sheet: /POWER/
File: power.kicad_sch

Title:

Size: A4
KiCad E.D.A. 9.0.9

Date: 2026-04-10

Rev: 0
Id: 5/5